

Cryptography

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Crypto

- ❑ **Cryptology** — The art and science of making and breaking “secret codes”
- ❑ **Cryptography** — making “secret codes”
- ❑ **Cryptanalysis** — breaking “secret codes”
- ❑ **Crypto** — all of the above (and more)

How to Speak Crypto

- ❑ A *cipher* or *cryptosystem* is used to *encrypt* the *plaintext*
- ❑ The result of encryption is *ciphertext*
- ❑ We *decrypt* ciphertext to recover plaintext
- ❑ A *key* is used to configure a cryptosystem
- ❑ A *symmetric key* cryptosystem uses the same key to encrypt as to decrypt
- ❑ A *public key* cryptosystem uses a *public key* to encrypt and a *private key* to decrypt

Crypto

❑ Basic assumptions

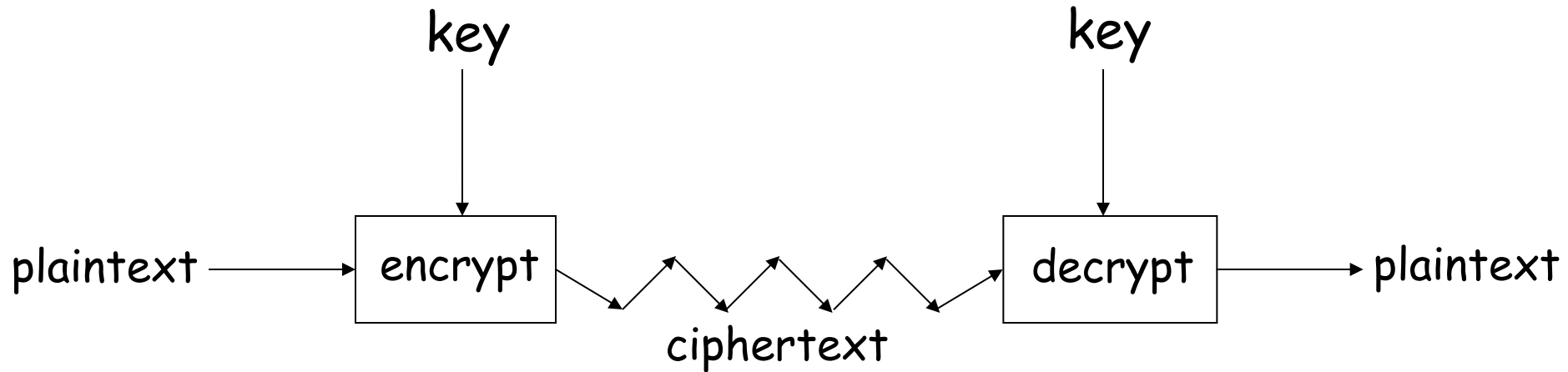
- The system is completely known to the attacker
- Only the key is secret
- That is, crypto algorithms are not secret

❑ This is known as **Kerckhoffs' Principle**

❑ Why do we make such an assumption?

- Experience has shown that secret algorithms tend to be weak when exposed
- Secret algorithms never remain secret
- Better to find weaknesses beforehand

Crypto as Black Box



A generic view of symmetric key crypto

Simple Substitution

❑ Plaintext: **fourscoreandsevenyearsago**

❑ Key:

Plaintext	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

❑ Ciphertext:

IRXUVFRUHDQGVHYHQBHDUVDJR

❑ Shift by 3 is "Caesar's cipher"

Ceasar's Cipher Decryption

- Suppose we know a Caesar's cipher is being used:

Plaintext	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

- Given ciphertext:

VSRQJHEREVTXDUHSDQWV

- Plaintext: spongebobsquarepants

Not-so-Simple Substitution

- Shift by n for some $n \in \{0, 1, 2, \dots, 25\}$
- Then key is n
- Example: key $n = 7$

Plaintext

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G

Ciphertext

Cryptanalysis I: Try Them All

- ❑ A simple substitution (shift by n) is used
 - But the key is unknown
- ❑ Given ciphertext: **CSYEVIXIVQMREXIH**
- ❑ How to find the key?
- ❑ Only 26 possible keys — try them all!
- ❑ **Exhaustive key search**
- ❑ Solution: key is $n = 4$

Simple Substitution: General Case

- ❑ In general, simple substitution key can be any **permutation** of letters
 - Not necessarily a shift of the alphabet
- ❑ For example

Plaintext	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext	J	I	C	A	X	S	E	Y	V	D	K	W	B	Q	T	Z	R	H	F	M	P	N	U	L	G	O

- ❑ Then $26! > 2^{88}$ possible keys

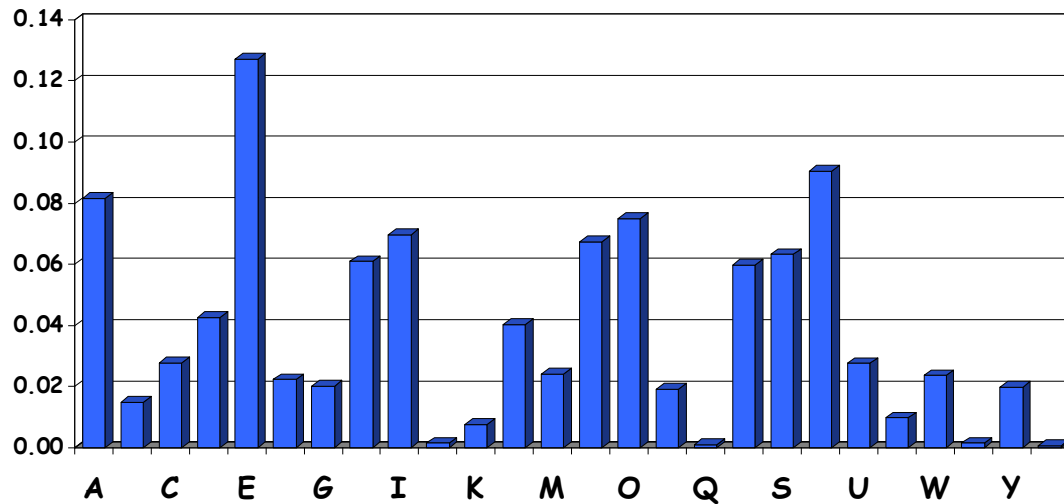
Cryptanalysis II: Be Clever

- ❑ We know that a simple substitution is used
- ❑ But not necessarily a shift by n
- ❑ Find the key given the ciphertext:

PBFPVYFBQXZTYFPBFEQJHDXXQVAPTPQJKTOYQWIPBVWLXTOX
BTFXQWAXBVCXQWAXFQJWVLEQNTQZQGGQLFXQWAKVWLXQ
WAEBIPBFXFQVXGTVJVWLBTPQWAEBFPBFHCVLXBQUFEVWLXGD
PEQVPQGVPPBFTIXPFHXZHVFAGFOTHFEFBQUFTDHzBQPOTHXTY
FTODXQHFTDPTOGHFQPBQWAQJJTODXQHFOQPWTBDHHIXQV
APBFZQHCFWPFHPBFIPBQWKFABVYYDZBOTHBPQPQJTQOTOGHF
QAPBFEQJHDXXQVAVXEBQPEFZBVFOJIWFFACFCCFHQWAUVWF
LQHGFVAFXQHfUFHILTtAVWAFFAWTEVOITDHFHFQAITIXPFH
XAFQHEFZQWGFLVWPTOFFA

Cryptanalysis II

- ❑ Cannot try all 2^{88} simple substitution keys
- ❑ Can we be more clever?
- ❑ English letter frequency counts...



Cryptanalysis II

□ Ciphertext:

PBFPVYFBQXZTYFPBFEQJHDXQVAPTPQJKTOYQWIPBVWLXTOXBTFXQ
WAXBVCXQWAXFQJWVLEQNTQZQGGQLFXQWAKVWLXQWAEBIPBFXFQ
VXGTVJVWLBTPQWAEFBFBFHCVLXBQUFEVWLXGDPEQVPQGVPPBFTIXPFH
XZHVFAGFOTHFEBQUFTDHzBQPOThXTYFTODXQHFTDPTOGHFQPBQW
AQJJTODXQHFOQPWTBDHHIXQVAPBFZQHCFWPFHPBFIQPBQWKFAVYY
DZBOTHQPBQPJTQOTOGHFQAPBFEQJHDXQVAVXEBQPEFZBVFOJIWFF
ACFCCFHQWAVVWFLQHGFVAFXQHUFHILTTAVWAFFAWTEVOITDHFH
FQAITIXPFHXAFQHEFZQWGFLVWPTOFFA

□ Analyze this message using statistics below

Ciphertext frequency counts:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
21	26	6	10	12	51	10	25	10	9	3	10	0	1	15	28	42	0	0	27	4	24	22	28	6	8

Frequency Analysis

1. **Collect and count** the frequency of letters in the ciphertext.
2. **Compare** the frequencies with known letter frequencies in the language (e.g., English).
3. **Make initial guesses** for the most frequent letters (e.g., $C \rightarrow E$, $T \rightarrow T$, $Z \rightarrow A$).
4. **Analyze common word patterns** (e.g., "THE", "AND", "IS") to refine mappings.
5. **Refine and adjust** based on the context and make corrections if needed.
6. **Finalize** the mappings and decrypt the entire message.

Notes on Frequency Analysis

- Frequency analysis is very powerful for monoalphabetic ciphers
 - but it's less effective with more complex ciphers, like polyalphabetic ciphers (e.g., Vigenère cipher).
- In real-world cases, ciphertext can have noise (random letters) or longer messages
 - requiring more careful adjustments, but the core method remains the same.

Cryptanalysis: Terminology

- ❑ Cryptosystem is **secure** if best know attack is to try all keys
 - Exhaustive key search, that is
- ❑ Cryptosystem is **insecure** if *any* shortcut attack is known
- ❑ But then insecure cipher might be harder to break than a secure cipher!

Vigenere Cipher

- ❑ Simple substitution is *monoalphabetic*
- ❑ Vigenere cipher is simple example of a *polyalphabetic* substitution
 - Caesars ciphers, based on a keyword
- ❑ For example, keyword CAT indicates shift by 2, shift by 0, shift by 19
 - Then repeat as needed

Vigenere Example

- Suppose that we want to encrypt
attackatdawn
- Encryption:

keyword:	CATCATCATCAT
plaintext:	attackatdawn
ciphertext:	ctmccdctwcwg
- Ciphertext is ctmccdctwcwg
- How to decrypt? How to attack?

Playfair ciphers

<i>p</i>	<i>l</i>	<i>a</i>	<i>y</i>	<i>f</i>
<i>i</i>	<i>r</i>	<i>b</i>	<i>c</i>	<i>d</i>
<i>e</i>	<i>g</i>	<i>h</i>	<i>k</i>	<i>m</i>
<i>n</i>	<i>o</i>	<i>q</i>	<i>s</i>	<i>t</i>
<i>u</i>	<i>v</i>	<i>w</i>	<i>x</i>	<i>z</i>

- ❑ polyalphabetic substitution cipher
 - encrypts digraphs (pairs of letters) rather than individual letters
- ❑ Matrix-based block cipher used in WWI
- ❑ In a 5x5 matrix, write the letters of the word "playfair" (for example) without dups, and fill in with other letters of the alphabet, except I,J used interchangeably.

Playfair encryption

<i>p</i>	<i>l</i>	<i>a</i>	<i>y</i>	<i>f</i>
<i>i</i>	<i>r</i>	<i>b</i>	<i>c</i>	<i>d</i>
<i>e</i>	<i>g</i>	<i>h</i>	<i>k</i>	<i>m</i>
<i>n</i>	<i>o</i>	<i>q</i>	<i>s</i>	<i>t</i>
<i>u</i>	<i>v</i>	<i>w</i>	<i>x</i>	<i>z</i>

He lx lo th er ex

KG YV RV QM GI KU

To decrypt, just reverse!

1. Break plaintext into letter pairs
 - o If a pair would contain double letters, split with x
 - o Pad end with x
 - o hellothere becomes...
 - o he lx lo th er ex
2. For each pair,
 - o If they are in the same row, replace each with the letter to its right (mod 5)
 - he → KG
 - o If they are in the same column, replace each with the letter below it (mod 5)
 - lo → RV
 - o Otherwise, replace each with letter we'd get if we swapped their column indices
 - lx → YV

Weaknesses

$$P = \begin{bmatrix} p & l & a & y & f \\ i & r & b & c & d \\ e & g & h & k & m \\ n & o & q & s & t \\ u & v & w & x & z \end{bmatrix}$$

- ❑ We have $26 * 26 = 676$ diagrams
- ❑ P is unknown, but structure of its bottom is predictable
- ❑ Can break using diagram frequencies and few hundred letters

Rail Fence Transposition

- ❑ Decide the number of rails (i.e., rows or depth of the matrix)
- ❑ Plaintext: This is a secret message
- ❑ Rail: 2

Rail Fence
Encoding

T		I		I		A		E		R		T		E		S		G	
	H		S		S		S		C		E		M		S		A		E

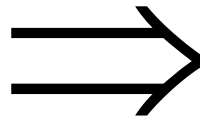
- ❑ Ciphertext: **TIIAERTESGHSSSCEMSAE**
- ❑ Key is the number of rails
- ❑ Different Rails number will result with different Ciphertext for the same plaintext

Double Transposition

□ Plaintext: **attackxatxdawn**

	col 1	col 2	col 3
row 1	a	t	t
row 2	a	c	k
row 3	x	a	t
row 4	x	d	a
row 5	w	n	x

Permute rows
and columns



	col 1	col 3	col 2
row 3	x	t	a
row 5	w	x	n
row 1	a	t	t
row 4	x	a	d
row 2	a	k	c

□ Ciphertext: **xtawxnattxadakc**

□ Key is matrix size and permutations:
(3,5,1,4,2) and (1,3,2)

One-Time Pad: Encryption

e=000 h=001 i=010 k=011 l=100 r=101 s=110 t=111

Encryption: Plaintext $\oplus$ Key = Ciphertext

	h	e	i	l	h	i	t	l	e	r
Plaintext:	001	000	010	100	001	010	111	100	000	101
Key:	111	101	110	101	111	100	000	101	110	000
Ciphertext:	110	101	100	001	110	110	111	001	110	101
	s	r	l	h	s	s	t	h	s	r

One-Time Pad: Decryption

e=000 h=001 i=010 k=011 l=100 r=101 s=110 t=111

Decryption: Ciphertext $\oplus$ Key = Plaintext

	s	r	l	h	s	s	t	h	s	r
Ciphertext:	110	101	100	001	110	110	111	001	110	101
Key:	111	101	110	101	111	100	000	101	110	000
Plaintext:	001	000	010	100	001	010	111	100	000	101
	h	e	i	l	h	i	t	l	e	r

One-Time Pad Summary

- ❑ **Provably** secure
 - Ciphertext gives **no** useful info about plaintext
 - All plaintexts are *equally likely*
- ❑ BUT, only when be used correctly
 - Pad must be random, used only once
 - Pad is known only to sender and receiver
- ❑ Note: pad (key) is same size as message

Real-World One-Time Pad

- ❑ Project VENONA
 - Soviet spies encrypted messages from U.S. to Moscow in 30's, 40's, and 50's
 - Nuclear espionage, etc.
 - Thousands of messages
- ❑ Spy carried one-time pad into U.S.
- ❑ Spy used pad to encrypt secret messages
- ❑ Repeats within the "one-time" pads made cryptanalysis possible

VENONA Decrypt (1944)

[C% Ruth] learned that her husband [v] was called up by the army but he was not sent to the front. He is a mechanical engineer and is now working at the ENORMOUS [ENORMOZ] [vi] plant in SANTA FE, New Mexico. [45 groups unrecoverable]

detain VOLOK [vii] who is working in a plant on ENORMOUS. He is a FELLOWCOUNTRYMAN [ZEMLYaK] [viii]. Yesterday he learned that they had dismissed him from his work. His active work in progressive organizations in the past was cause of his dismissal. In the FELLOWCOUNTRYMAN line LIBERAL is in touch with CHESTER [ix]. They meet once a month for the payment of dues. CHESTER is interested in whether we are satisfied with the collaboration and whether there are not any misunderstandings. He does not inquire about specific items of work [KONKRETNAYa RABOTA]. In as much as CHESTER knows about the role of LIBERAL's group we beg consent to ask C. through LIBERAL about leads from among people who are working on ENOURMOUS and in other technical fields.

- ❑ "Ruth" == Ruth Greenglass
- ❑ "Liberal" == Julius Rosenberg
- ❑ "Enormous" == the atomic bomb

Codebook Cipher

- ❑ Literally, a book filled with “codewords”
- ❑ Zimmerman Telegram encrypted via codebook

Februar	13605
fest	13732
finanzielle	13850
folgender	13918
Frieden	17142
Friedenschluss	17149
:	:

- ❑ Modern block ciphers are codebooks!
- ❑ More about this later...

Codebook Cipher: Additive

- ❑ Codebooks also (usually) use **additive**
- ❑ Additive — book of “random” numbers
 - Encrypt message with codebook
 - Then choose position in additive book
 - Add in additives to get ciphertext
 - Send ciphertext and additive position (MI)
 - Recipient subtracts additives before decrypting

Zimmerman Telegram

- Perhaps most famous codebook ciphertext ever
- A major factor in U.S. entry into World War I

WESTERN UNION TELEGRAM

CLASS OF SERVICE DESIRED
☒ Post Day Message
☐ Day Letter
☐ Night Message
☐ Night Letter

Persons should make up their mind as to what they wish to send, and if they wish to send a message, they should send it. OTHERWISE THE TELEGRAM WILL BE TRANSMITTED AS A FAST DAY MESSAGE.

Send the following telegram, subject to the terms on back hereof, which are hereby agreed to

via Galveston

JAN 19 1917

GERMAN LEGATION
MEXICO CITY

130	13042	13401	8501	115	3528	416	17214	8491	11310
18147	18222	21560	10247	11518	23677	13605	3494	14936	
98092	5905	11311	10392	10371	0302	21290	5161	39695	
23571	17504	11269	18276	18101	0317	0228	17694	4473	
22284	22200	19452	21589	87893	5569	13918	8958	12137	
1333	4725	4458	5905	17166	13851	4458	17149	14471	6708
13850	12224	6929	14991	7382	15857	67893	14218	36477	
5870	17553	67893	5870	5454	16102	15217	22801	17138	
21001	17388	7446	23638	18222	6719	14331	15021	23845	
3156	23552	22096	21604	4797	9497	22464	20855	4377	
23610	18140	22260	5905	13347	20420	39689	13732	20667	
6929	5275	18507	52262	1340	22049	13339	11265	22295	
10439	14814	4178	6992	8784	7032	7357	6926	52262	11267
21100	21272	9346	9559	22464	15874	18502	18500	15857	
2188	5376	7381	98092	16127	13486	9350	9220	76036	14219
5144	2831	17920	11347	17142	11264	7667	7762	15099	9110
10482	97556	3569	3670						

BEPNSTORFF.

Charge German Embassy.

Zimmerman Telegram Decrypted

- ❑ British had recovered partial codebook
- ❑ Then able to fill in missing parts

MAILED
October 1-8-18
Washington, State Dept.
By *Wm. A. Eckhoff*
Date *Oct. 27, 1918*

TELEGRAM RECEIVED.

FROM 2nd from London # 5747.

"We intend to begin on the first of February unrestricted submarine warfare. We shall endeavor in spite of this to keep the United States of America neutral. In the event of this not succeeding, we make Mexico a proposal of alliance on the following basis: make war together, make peace together, generous financial support and an understanding on our part that Mexico is to reconquer the lost territory in Texas, New Mexico, and Arizona. The settlement in detail is left to you. You will inform the President of the above most secretly as soon as the outbreak of war with the United States of America is certain and add the suggestion that he should, on his own initiative, ~~invite~~ ^{invite} Japan to immediate adherence and at the same time mediate between Japan and ourselves. Please call the President's attention to the fact that the ruthless employment of our submarines now offers the prospect of compelling England in a few months to make peace." Signed, ZIMMERMAN.

Election of 1876

- ❑ Apply codebook to original message
- ❑ Pad message to multiple of 5 words (total length, 10,15,20,25 or 30 words)
- ❑ For each length, a fixed permutation applied to resulting message
- ❑ Permutations found by comparing several messages of same length
- ❑ Note that the **same key** is applied to all messages of a given length

Election of 1876

- ❑ Ciphertext: **Warsaw they read all unchanged last are idiots can't situation**
- ❑ Codebook: Warsaw == telegram
- ❑ Transposition: 9,3,6,1,10,5,2,7,4,8
- ❑ Plaintext: **Can't read last telegram. Situation unchanged. They are all idiots.**
- ❑ A weak cipher made worse by reuse of key
- ❑ Lesson? Don't overuse keys!

Early 20th Century

- ❑ WWI — Zimmerman Telegram
- ❑ "Gentlemen do not read each other's mail"
 - Henry L. Stimson, Secretary of State, 1929
- ❑ WWII — **golden** age of cryptanalysis
 - Midway/Coral Sea
 - Japanese **Purple** (codename MAGIC)
 - German **Enigma** (codename ULTRA)

Post-WWII History

- ❑ Claude Shannon — father of the science of information theory
- ❑ Computer revolution — lots of data to protect
- ❑ Data Encryption Standard (DES), 70's
- ❑ Public Key cryptography, 70's
- ❑ CRYPTO conferences, 80's
- ❑ Advanced Encryption Standard (AES), 90's

Claude Shannon

- ❑ The founder of Information Theory
- ❑ 1949 paper: [Comm. Thy. of Secrecy Systems](#)
- ❑ Fundamental concepts
 - **Confusion** — obscure relationship between plaintext and ciphertext
 - **Diffusion** — spread plaintext statistics through the ciphertext
- ❑ Proved one-time pad is secure
- ❑ One-time pad is confusion-only, while double transposition is diffusion-only

Taxonomy of Cryptography

❑ Symmetric Key

- Same key for encryption and decryption
- Modern types: Stream ciphers, Block ciphers

❑ Public Key (or "asymmetric" crypto)

- Two keys, one for encryption (public), and one for decryption (private)
- And digital signatures — nothing comparable in symmetric key crypto

❑ Hash algorithms

- Can be viewed as "one way" crypto